

Merrivoss Bioanalysis Inc, a Tarnovell company

OB11 GPCR antagonists: five-isoform CYP inhibition at screening concentration

Final Report

Study MV-2022-176 (LIMS M22/BIO/04063)

Client: Orvessant Biosciences Pty Ltd

Study Director: Dr M. Larkin

Report date: August 10, 2022

1 Summary

40 compounds from the Orvessant piperidyl benzamide series 11 were tested for Orvessant Biosciences Pty Ltd in the Five-isoform DDI screen. Testing started on July 25, 2022 and results were reported on August 10, 2022. All runs met matrix acceptance criteria. Inhibition above 50% at 10 μ M was most frequent for CYP2C9.

Table 1 Study summary

Metric	Value
Project	MV-2022-176
Client	Orvessant Biosciences Pty Ltd
Compounds tested	40
Series / chemotype	Orvessant piperidyl benzamide series 11 (Piperidyl benzamide)
Assays	CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	16
Price	USD 12,400
CYP1A2 >50% inhibition @10 μ M	10
CYP2C9 >50% inhibition @10 μ M	12
CYP2C19 >50% inhibition @10 μ M	11
CYP2D6 >50% inhibition @10 μ M	11
CYP3A4 >50% inhibition @10 μ M	11
Compounds with any CYP >50%	18
Ketoconazole (3A4) IC50 (μ M)	0.031

2 Objective

To rank the test article set for Five-isoform DDI screen liability so that Orvessant Biosciences Pty Ltd could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on July 25, 2022 and logged against M22/BIO/04063. Identity and concentration were taken as supplied by the client.

4 Methods

Five-isoform DDI screen

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation was measured by LC-MS/MS against vehicle (0.1% DMSO) incubations.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Results are reported as mean percent inhibition relative to vehicle. Values above 50% at 10 μ M are flagged for follow-up; small negative values indicate no inhibition.

5 Results

CYP inhibition results

Inhibition above 50% at 10 μ M was most frequent for CYP2C9. In total 18 compounds inhibited at least one isoform by more than 50% at 10 μ M (CYP3A4: 11; CYP2D6: 11; CYP2C19: 11; CYP2C9: 12; CYP1A2: 10). Ketoconazole inhibited CYP3A4 with an IC_{50} of 0.031 μ M.

Table 3 Percent inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
OB11-08904	12	12	-9	6	2
OB11-08906	16	56	6	20	34
OB11-08910	7	4	43	3	4
OB11-08916	8	-8	11	59	37
OB11-08918	65	93	93	7	95
OB11-08922	20	54	41	96	34
OB11-08928	3	8	19	10	2
OB11-08930	4	11	1	10	-1
OB11-08934	-4	8	5	5	-7
OB11-08936	20	23	73	60	54
OB11-08942	1	13	33	1	-4
OB11-08948	1	77	70	3	56
OB11-08952	9	2	12	23	-5
OB11-08964	12	4	35	10	-3
OB11-08970	88	91	100	5	44

OB11-08972	87	79	67	21	76
OB11-08978	-0	3	13	92	-3
OB11-08982	-12	1	3	5	2
OB11-08984	-1	5	2	1	-7
OB11-08990	58	95	100	88	65
OB11-08994	-4	4	1	13	-4
OB11-08996	-5	20	7	18	-9
OB11-09000	9	3	3	21	1
OB11-09008	94	93	92	75	100
OB11-09012	2	6	23	8	-1
OB11-09026	9	2	27	0	14
OB11-09030	6	3	9	1	9
OB11-09032	96	89	94	33	97
OB11-09036	7	10	38	-4	5
OB11-09042	2	-0	1	4	-0
OB11-09044	6	-1	-2	-12	-1
OB11-09048	10	2	33	5	3
OB11-09054	59	40	9	78	66
OB11-09060	9	7	18	97	26
OB11-09062	9	8	4	75	13
OB11-09066	100	100	92	96	87
OB11-09074	6	-3	6	6	9
OB11-09080	83	96	89	45	85
OB11-09086	100	88	67	72	80
OB11-09090	6	-4	5	13	9

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.45; ticlopidine (CYP2C19) 1.2; quinidine (CYP2D6) 0.064; ketoconazole (CYP3A4) 0.031

6 Discussion

Inhibition above 50% at 10 μ M was most frequent for CYP2C9. Inhibition values are in vitro, at a fixed concentration, and do not by themselves predict clinical interaction risk; free exposure estimates are needed for that assessment.

7 Conclusion

The data support ranking within the set. Recommended next step: carry the compounds without CYP flags into exposure studies and run CYP3A4 IC₅₀s for flagged analogues before in vivo work.

Name	Role	Signature	Date
Megan Larkin	Study Director	<i>Larkin</i>	August 10, 2022
Helen Marchwood	Bioanalytical Quality Manager	<i>Marchwood</i>	August 9, 2022